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SEQUOYAH N. WALTERS

EDUCATION

Doctoral Student (Ph.D.), Mechanical Engineering | September 2020 – present

- ♦ University of Wisconsin, Madison

B.S., Physics | Minors, Comp Sci and Math | August 2015 – December 2019

- ♦ West Chester University Honors College

WORK EXPERIENCE

Mech. Eng. Research Assistant 6/2020 – present

- ♦ Under mentorship of Dr. Xiangru Xu, research includes the design and validation of control algorithms to guarantee safety and performance for autonomous systems.

Physics Research Assistant (Morphology of Aerosols) 1/2020 – 6/2020

- ♦ Simulation research using MATLAB for characterizing the morphology of aerosol particles. Work is mentored by Dr. Aptowicz of WCU. Manuscript in progress.

Physics Tutor 8/2018 – 12/2019

- ♦ Algebra-based Physics I and II with CRLA certification.

Mechanical Engineering Research Assistant 5/2019 – 8/2019

- ♦ Competitive summer research program held at the University of Wisconsin, Madison
- ♦ Calculated knee moment data for trans-tibial amputees walking on slopes and stairs using the Variable - Stiffness Foot prosthesis, under mentorship of Dr. Adamczyk
- ♦ **Led a robotics team** of high school students in building and testing multiple small-scale robotics projects, including two mobile robots and a robotic arm.

Physics Research Assistant 5/2017 – 8/2017

- ♦ Aided WCU professor Dr. Aptowicz in aerosol absorption simulation research utilizing MATLAB and Fortran code. See “publication” section for the article.

VOLUNTEER EXPERIENCE

Project Lead - Operation Eco Garden Building 5/2018 – 12/2019

- ♦ Turning wooden pallets into gardens for food insecure families

Rebuilding Damaged Houses in Windsor, NC 1/2017, 1/2018

- ♦ Two one-week volunteer trips in response to hurricane Matthews

New Directions After School Tutoring Program 10/2016 – 3/2017

- ♦ Tutored elementary students on all topics and helped with homework

STUDENT INVOLVEMENT

Friars' Society 3/2017 – 12/2019

- ♦ Service based organization for volunteering in numerous community service projects

Founder and President of the WCU Judo Club 1/2016 – 2/2018

- ♦ Oversaw all judo club functions including practices and home tournaments
- ♦ Maintained and encouraged clear communication and motivation with all members

Society of Physics Students 9/2015 – 12/2019

- ♦ Organization for growing relationships between physics students outside of academia

AWARDS, ACHIEVEMENTS

GERS Fellowship – 9/2020

- ♦ Graduate Engineering Research Scholars fellowship to fund 3 years of graduate study.

College of the Sciences and Mathematics Outstanding Student Award – 12/2019

- ♦ Recognizes one senior student who has shown exceptional intellectual or creative achievement and involvement in extracurricular and service activities.

Student Commencement Speaker College of the Sciences and Mathematics – 12/2019

Dr. Russel K. Rickert Award for Research Excellence – 4/2019

2018 State APSCUF Scholarship – 8/2018

- ♦ Awarded for academic merit and volunteerism to two students from the 14 Pennsylvania state-owned universities who are relatives of an APSCUF member

APSCUF Labor Union Scholarship – 6/2017

- ♦ Awarded for academic achievement and leadership characteristics to a student at WCU who is a relative of a recognized labor-union member

The Roger and Diane Casagrande Scholarship – 3/2017

- ♦ Awarded for academic achievement and diligence in the field of physics

W.W. Smith Scholarship – 8/2015 – 8/2018

- ♦ A yearly scholarship awarded for academic achievement and financial need

PUBLICATION, POSTER PRESENTATIONS, CONFERENCES

Publication

- ♦ “Characterizing the size and absorption of single nonspherical aerosol particles from angularly-resolved elastic light scattering,” Sequoyah Walters, Jason Zallie, Gabe Seymour, Yong-Le Pan, Gorden Videen, Kevin B. Aptowicz, Journal of Quantitative Spectroscopy and Radiative Transfer. (published).

Mechanical Engineering REU Poster Presentation

- ♦ Presented poster “Slopes and Stairs: Knee Moments with the Variable-Stiffness Foot”
- ♦ University of Wisconsin, Madison – July 31, 2019

17th Electromagnetic and Light Scattering Conference (presented poster)

- ♦ Presented on characterizing absorption of single-particle aerosols from TAOS scattering patterns for NASA/Texas A&M University-funded conference
- ♦ Texas A&M University – March 4-9, 2018

SKILLS, PROJECTS

Software/Programming Skills

- ♦ C++, Java, Linux, MATLAB, Python, Robot Operating System (ROS), CAD

Technical Hardware Skills

- ♦ Raspberry Pi, Arduino, OptiTrack Motion Capture, i2c device usage, 3D printing

Coursework

- ♦ Electronics, Mechanics, Optics, Thermodynamics, Quantum Mechanics, Linear Algebra, Applied Partial Differential Equations, Data Structures and Algorithms

Projects

- ♦ Differential drive monocular SLAM (Simultaneous Localization and Mapping) robot using a Raspberry Pi and a server. The software framework is ROS, written in C++. The chassis and many parts were designed in OnShape.
- ♦ Two-wheel self-balancing robot using IMU and Arduino, controlled using a PID control algorithm. Most structural parts were designed with OnShape and 3D printed.